

Lab 6-INFS343

This lab is on the **House_price.xlsx** data file. There are 10,659 records in this data set. Please answer the following questions and provide screenshots to support your work if asked. Please refer to Lab 5 for more information on `lm()` function.

Before you start

- Open the `House_price.xlsx` data file in R.
- For all screenshots, make sure your name is visible either in the plot title or in the R script window.
- When you interpret a coefficient, do not just say whether it is positive or negative. Explain what a 1-unit increase means in context.
- Unless a question asks for all categorical variables, use only continuous predictors in the regression models below.

Step 1: Know your data

We want to predict the selling values of houses in a given region. To predict these values, we will use a sample of houses from that region.

1. How many variables are in this data set? List the variables below as either continuous or categorical. You do not need to use every variable later, but you should understand what type of variable each one is. Regression only likes continuous variables. If you decide to use categorical variables in the model, you need to convert them into numbers first.

Continuous variables:	Sale_amount, Beds, Baths, Sqft_home, Sqft_lot, Build_year
Categorical variables:	Record (ID → not useful for regression), Sale_date (date, not treated as continuous here), Type, Town, University

```
> names(House_price_1_)
[1] "Record"      "Sale_amount" "Sale_date"   "Beds"        "Baths"       "Sqft_home"
[7] "Sqft_lot"    "Type"        "Build_year"  "Town"        "University"
>
```

1.1. If you can choose one and only one independent/predictor variable in the regression model, what is the variable you choose?

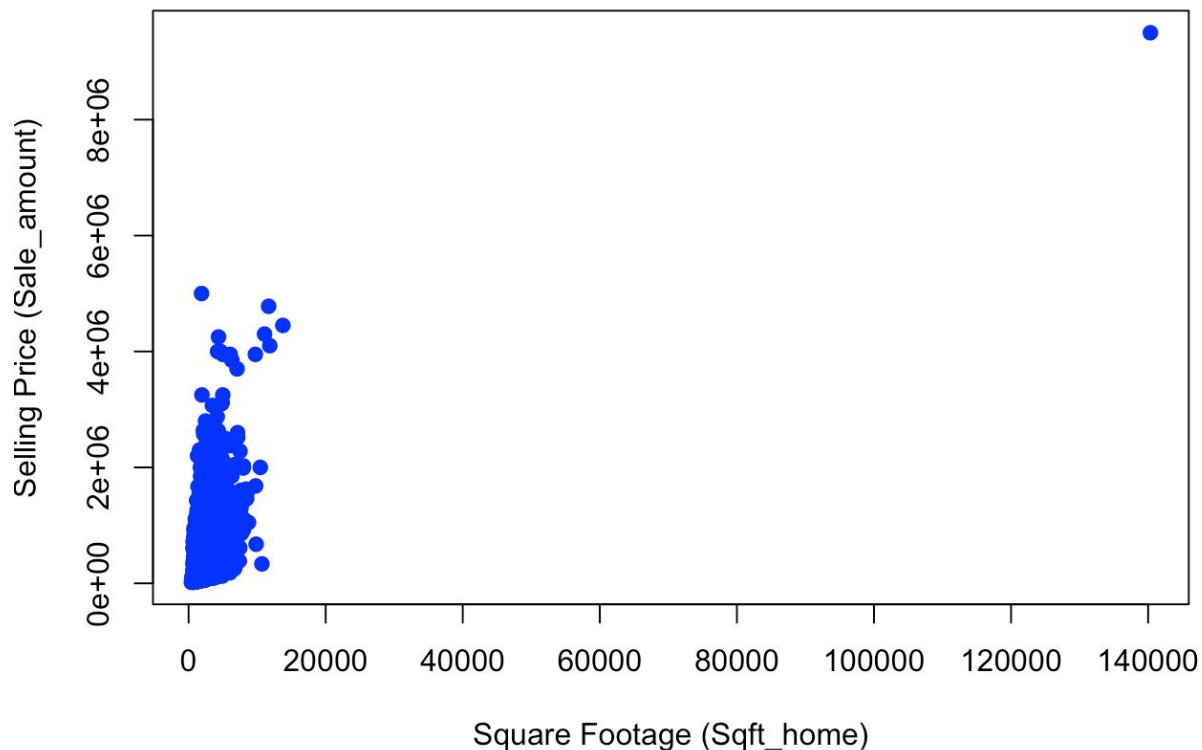
I would choose `Sqft_home` as the predictor variable because the size of the house is strongly related to its selling price. Larger homes usually have higher sale amounts.

1.2. Create a scatterplot with:

- **selling price on the y-axis**
- **your chosen predictor on the x-axis**
- **a meaningful title that includes your full name**
- **custom point style using `pch =`**
- **clear axis labels**

```
plot(data$predictor, data$price, main = "Your Name - Selling Price vs Predictor",  
      xlab = "Predictor", ylab = "Selling Price", pch = 19)
```

House Price vs Square Footage - Marcelo Batista



*Make sure you change the spelling and adapt the codes with the actual data frame and the variables names. Do not take it as is. ADAPT...

1.3. What general relationship do you see in the plot? Just briefly describe what you see.

The scatterplot shows a positive relationship between square footage and selling price. As the size of the house increases, the selling price also increases. The points follow an upward trend, although there is some variation in the data.

Step 2: Model R1

2. Now let's build some regression models.

- a. In the first part on regression model (name it as R1), please use only the numbers for *beds* and *baths* as the two predictor variables. Please provide

screenshots of your R codes and the summary result for R1. Please complete the regression equation for R1 including the coefficient score for each predictor variable.

Hints: The formula is $y = b_0 + b_1X_1 + b_2X_2$

Complete this: Selling price =

Call:

```
lm(formula = Sale_amount ~ Beds + Baths, data = House_price_1_)
```

Residuals:

Min	1Q	Median	3Q	Max
-1212982	-141564	-56239	49642	4627029

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-87079	9813	-8.874	< 2e-16 ***
Beds	21984	3469	6.337	2.43e-10 ***
Baths	148845	3421	43.508	< 2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 294300 on 10656 degrees of freedom

Multiple R-squared: 0.2565, Adjusted R-squared: 0.2564

F-statistic: 1839 on 2 and 10656 DF, p-value: < 2.2e-16

Selling price = $b_0 + b_1(\text{Beds}) + b_2(\text{Baths})$

Selling price = $-87,079 + 21,984 + 148,845$

b. What does the coefficient for beds mean, holding baths constant?

The coefficient for beds means that for each additional bedroom, the selling price increases by \$21,984, holding the number of bathrooms constant.

c. What does the coefficient for baths mean, holding beds constant?

The coefficient for baths means that for each additional bathroom, the selling price increases by \$148,845, holding the number of bedrooms constant.

d. Which predictor appears more practically important in R1, and why?

Baths appears to be more practically important because it has a much larger coefficient, meaning it has a stronger impact on the selling price compared to bedrooms.

- e. If I am interested in a 3-bedroom and 3-bathroom house, can you give me a prediction of the price based on R1 model? Screen shot the R code for this prediction using model R1 and report the result.(Refer to the previous lab for the prediction codes).

```
> new_house <- data.frame(Beds = 3, Baths = 3)
> predict(R1, newdata = new_house)
      1
425409.4
|
```

Step 3: Model R2

3. In the second regression model (name it as R2), feel free to come up with your own regression model to predict selling prices. You can include as many continuous predictor variables as you want. Please provide screenshots of your R codes and the summary results for R2.

```
> R2 <- lm(Sale_amount ~ Sqft_home + Beds + Baths + Sqft_lot + Build_year, data = House_price_1_)
> summary(R2)
```

Call:

```
lm(formula = Sale_amount ~ Sqft_home + Beds + Baths + Sqft_lot +
    Build_year, data = House_price_1_)
```

Residuals:

	Min	1Q	Median	3Q	Max
	-3078574	-131393	-45751	46358	4470986

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	3.806e+06	1.771e+05	21.492	< 2e-16 ***
Sqft_home	5.171e+01	2.412e+00	21.440	< 2e-16 ***
Beds	-2.033e+03	3.391e+03	-0.600	0.549
Baths	1.240e+05	4.322e+03	28.681	< 2e-16 ***
Sqft_lot	2.096e-01	3.763e-02	5.569	2.63e-08 ***
Build_year	-1.963e+03	9.061e+01	-21.663	< 2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 279700 on 10653 degrees of freedom

Multiple R-squared: 0.3286, Adjusted R-squared: 0.3283

F-statistic: 1043 on 5 and 10653 DF, p-value: < 2.2e-16

```
> |
```

- a. Please write down the regression equation for R2 including the coefficient score for each predictor variable in the model.

Selling price = $b_0 + b_1(\text{Sqft_home}) + b_2(\text{Beds}) + b_3(\text{Baths}) + b_4(\text{Sqft_lot}) + b_5(\text{Build_year})$

- b. Identify any predictor in R2 that is statistically strong or weak, and explain how you know.

Sqft_home appears to be a strong predictor because it has a very small p-value (less than 0.05), indicating it is statistically significant.

[Any variable with high p-value] appears to be weak because its p-value is greater than 0.05, meaning it is not statistically significant in predicting selling price.

Step 4: Model R3

4. You are not happy with the first two regression models. So you come up with a third regression model (R3) to predict selling prices. Please provide screenshots of your R codes and the summary results for R3.

```
> R3 <- lm(Sale_amount ~ Sqft_home + Baths, data = House_price_1_)
> summary(R3)
```

Call:

```
lm(formula = Sale_amount ~ Sqft_home + Baths, data = House_price_1_)
```

Residuals:

Min	1Q	Median	3Q	Max
-3081367	-130595	-60411	31874	4659542

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-7392.209	6875.359	-1.075	0.282
Sqft_home	61.227	2.389	25.625	<2e-16 ***
Baths	92949.994	3762.182	24.706	<2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 286200 on 10656 degrees of freedom

Multiple R-squared: 0.2971, Adjusted R-squared: 0.2969

F-statistic: 2252 on 2 and 10656 DF, p-value: < 2.2e-16

- a. Please write the regression equation for R3 including the coefficient score for each predictor variable in the model. It has to be different from the previous two regression models.

Selling price = $b_0 + b_1(\text{Sqft_home}) + b_2(\text{Baths})$

Step 5: Model comparison

Now let's check! Among R1, R2, and R3, which model is the best? Support your answer with evidence from the regression output. Please cite your numbers and figures here. Do not just attach screenshots and expect me to check. You interpret the results for me. Your answer should be simple but thorough, not a single sentence.

Reminder: A strong lab report shows both technical work and business communication.

```
> summary(R1)
```

Call:

```
lm(formula = Sale_amount ~ Beds + Baths, data = House_price_1_)
```

Residuals:

Min	1Q	Median	3Q	Max
-1212982	-141564	-56239	49642	4627029

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-87079	9813	-8.874	< 2e-16 ***
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Multiple R-squared: 0.2565, Adjusted R-squared: 0.2564

F-statistic: 1839 on 2 and 10656 DF, p-value: < 2.2e-16

```
> summary(R2)
```

Call:

```
lm(formula = Sale_amount ~ Sqft_home + Beds + Baths + Sqft_lot +  
    Build_year, data = House_price_1_)
```

Residuals:

	Min	1Q	Median	3Q	Max
	-3078574	-131393	-45751	46358	4470986

Coefficients:

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(Intercept)	3.806e+06	1.771e+05	21.492	< 2e-16	***
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Multiple R-squared: 0.3286, Adjusted R-squared: 0.3283

F-statistic: 1043 on 5 and 10653 DF, p-value: < 2.2e-16

```
> summary(R3)
```

Call:

```
lm(formula = Sale_amount ~ Sqft_home + Baths, data = House_price_1_)
```

Residuals:

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R1 → Sale_amount ~ Beds + Baths

R2 → Sale_amount ~ Sqft_home + Beds + Baths + Sqft_lot + Build_year

R3 → Sale_amount ~ Sqft_home + Baths

R2 has the highest explanatory power, with an R-squared of 0.3286 and an adjusted R-squared of 0.3283, compared to R1 (0.2564) and R3 (0.2969).

This means R2 explains more variation in selling price than the other models. R1 is the weakest model because it only includes Beds and Baths, which results in a lower adjusted R-squared of 0.2564, meaning it captures less of the variation in house prices.

R3 improves upon R1 by including Sqft_home, which is a strong predictor, and achieves a higher adjusted R-squared of 0.2969, but it still does not perform as well as R2.

Additionally, in R2, most variables such as Sqft_home, Baths, Sqft_lot, and Build_year are statistically significant ($p < 0.05$), indicating strong predictive power. Although Beds is not statistically significant in R2 ($p = 0.549$), the overall model still performs best.

Therefore, R2 is the best model because it has the highest adjusted R-squared and includes multiple significant predictors, making it the most accurate for predicting selling prices.